

Advanced Autonomous Systems

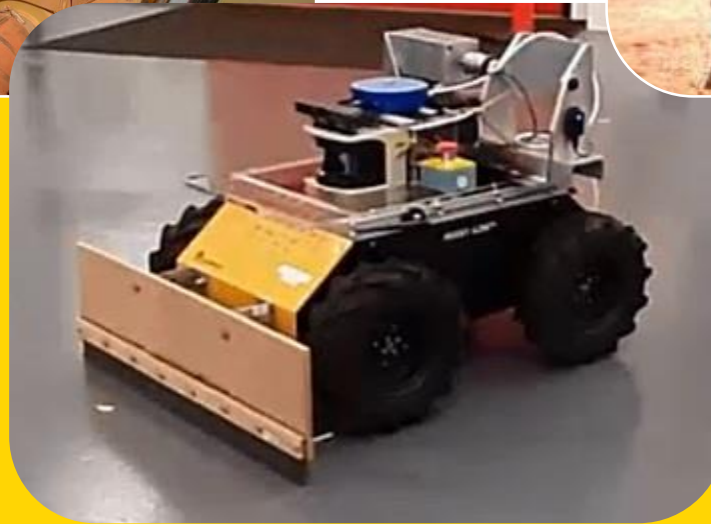
MTRN 4010



Lecturer:
**Jose
Guivant**



UNSW
AUSTRALIA



MTRN 4010



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2023 TERM1

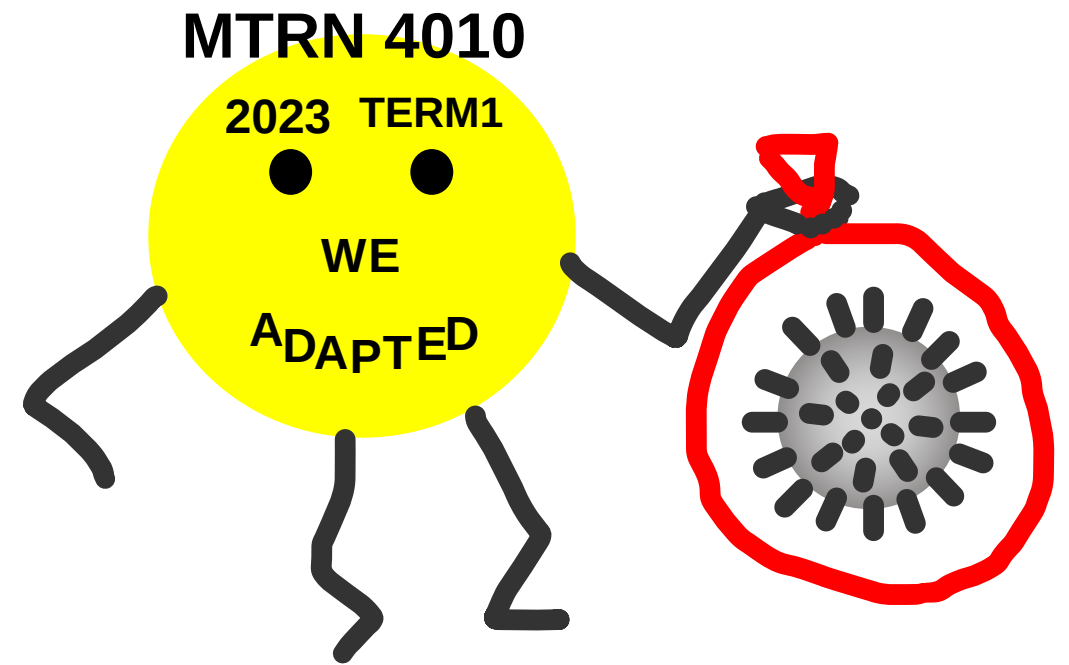
Lectures

→ F2F and Online

→ 2 hours

**Practical
component**
tutorial sessions project work

→
in person
via Teams



Teaching Staff

Lecturer: Jose Guivant j.guivant@unsw.edu.au

Leader demonstrators:

Ming Xuan Chua (Calvin)

Ahmed Ahmed

Demonstrators

Zhiyang Lin (Kilam)

Mustafa Oltan Sevinc

Max Keller

Diwen Ye (Hayden)

Hong Liang

Our leader Tutors are advanced PhD students

Ahmed Ahmed has almost completed his PhD thesis. He has more than 10 years of industrial experience in control engineering. His research is focused on nonlinear and robust control of over-actuated mechatronic systems such as certain class of powerful aerial drones.

Ming Xuan Chua is a PhD student, working on developing upper limb exoskeleton for rehabilitation.

Our tutors are all advanced UG students, who have excellent knowledge about MTRN4010 topic.

Lectures (Simultaneously via Teams and F2F)

Our lectures will be of 2 hours, nominally. We have an extra ½ hour for cases in which we need to extend our discussions, related to the theory or to the projects. Quizzes will take place at the beginning of the lecture time.

Practical Component

The practical sessions will take place in the **lab J18/212** (F2F sessions) and via **Teams** (online sessions)
The practical component will be composed by tutorial sessions and by project work.

Lecture content

Week 1:

Lecture 1 **Refreshing necessary concepts**

Vectors, matrixes, transformations. Coordinate frames. Rotation matrix.
State space models for continuous time systems.
LTI systems, Linear time variant systems, nonlinear systems.
Approximate discrete time models. Euler approximation, for non LTI cases.
Disturbances.
Classic Observers (continuous time, for LTI systems.)

Week 2:

Lecture 2. Part 1:

Usual Models in Field Robotics:

Analog and discrete time kinematic models of a wheeled platform.

Simulating models. Using models for prediction.

Limitations of deterministic predictions. Discrepancy due to uncertainty and disturbances.

Lecture 2. Part 2

SENSORS, part 1: IMU (Inertial measurement Unit).

Description. Relevant technical specifications.

Example using real IMUs. We inspect usual measurements, from IMUs operating in real time

Lecture 2. Part 3

Sensor Data Fusion(Part 1)

Combining models and sensors' measurements.

IMU integration for estimating 2D and 3D attitude.

Week 3:

Lecture 3. Part 1:

SENSORS (part 2). LiDARs

Description, typical data.

We see /inspect sensors working in class.

Lecture 3. Part 2:

Refreshing concepts about Statistics

Random variable (RV)

Probability Density Function (PDF) .

Multivariate PDFs in 2D and n-dimensions.

Gaussian cases. Expected value and covariance Matrix

Probability Mass.

Marginal and Conditional PDFs.

Statistical independence. White noise.

Bayes Rule.

Release of Project 1. Explanation.

Week 4

Transformation of Random variables (RVs.)

Linear and nonlinear transformations of Gaussian RVs.

Approximation via Linearization.

Bayesian fusion of multiple sources of information.

State estimation of dynamic systems. Sensor Data Fusion (part 1)

Week 5

Bayesian Estimation of dynamic systems. Sensor Data Fusion (part 2)

Gaussian Estimators.

Kalman Filter (KF),

Extended Kalman Filter (EKF).

Examples using KF and EKF.

Week 7

EKF (continuation).

Robot Localization. Dead Reckoning. Map based localization.

Sensor data fusion for Localization (case using IMU + encoder + LiDAR)

Deterministic approach. Bayesian approach. EKF based approach.

Using EKF for real-time model parameter identification (General concept)

Exploiting approach for refining the process model in localization.

Release of Project 2. Introduction to Project 2.

Week 8

Lecture, part 1

Estimating model parameters (general cases). Estimating platform's model parameters (in runtime, via EKF)

Lecture, part 2

Fusing platform's pose estimates and LiDAR measurements for mapping.
Occupancy grids.

Week 9

Optimization (common methods)

What is an optimizer?

Basic localization via optimization

Examples/cases: estimating model parameters (e.g., UGV kinematic model and pendulum model)

Particle Swarm Optimization (PSO)

Week 10

Bellman Principle of Optimality

Dynamic Programming approach

Dijkstra method.

Path Planning.

Assessment

2 Projects (2%,of total mark, each.)

3 Minor Quizzes (2% each)

1 Major Quiz (16%)

Final Exam (30%)

Project 1: submission deadline [Week 7, Tuesday, 11:55PM] (*)

Project 2: submission deadline [Week 10, Friday, 11:55PM] (**)

Minor quiz 1: Week 2, dt=10 minutes, during lecture time, Moodle.

Minor quiz 2: Week 4, dt=10 minutes, during lecture time, Moodle.

Minor quiz 3: Week 7, dt=10 minutes, during lecture time, Moodle.

Major quiz : Week 8, dt=35 minutes, during lecture time, Moodle.

Final Exam: (exam period).

(*) Submission + demonstration (which may include a quiz about the project.)

(**) Submission open in week 9. (you are allowed to re-submit it)

Assessment / Projects

2 Projects (23%+25% of total mark)

Submission + demonstration (which may include a quiz about the submitted solution)

Demonstrations: you show your implementation/solution working.

Evaluator may ask you to :

- 1) show, working, particular parts (multiple times)
- 2) show your implementation (e.g. ,source code)
- 3) ask for explanation of parts of your implementation
- 4) basic questions related to the project.

We may replace (4) by a general quiz, so that everyone is exposed to the same questions. If we proceed in that way, that quiz will take place during lecture time. You will be informed one week in advance.

Course Evaluation and Development

Lecture time is now also dedicated to discussing the application of EKF to systems in other areas of Engineering (in addition to the localization problem in the Robotics field)

Week 1 covers a review of concepts which are necessary for dealing with the course's new concepts. These reviewed concepts used to be assumed as well known, however are now reviewed to help students who may not be familiar with them.

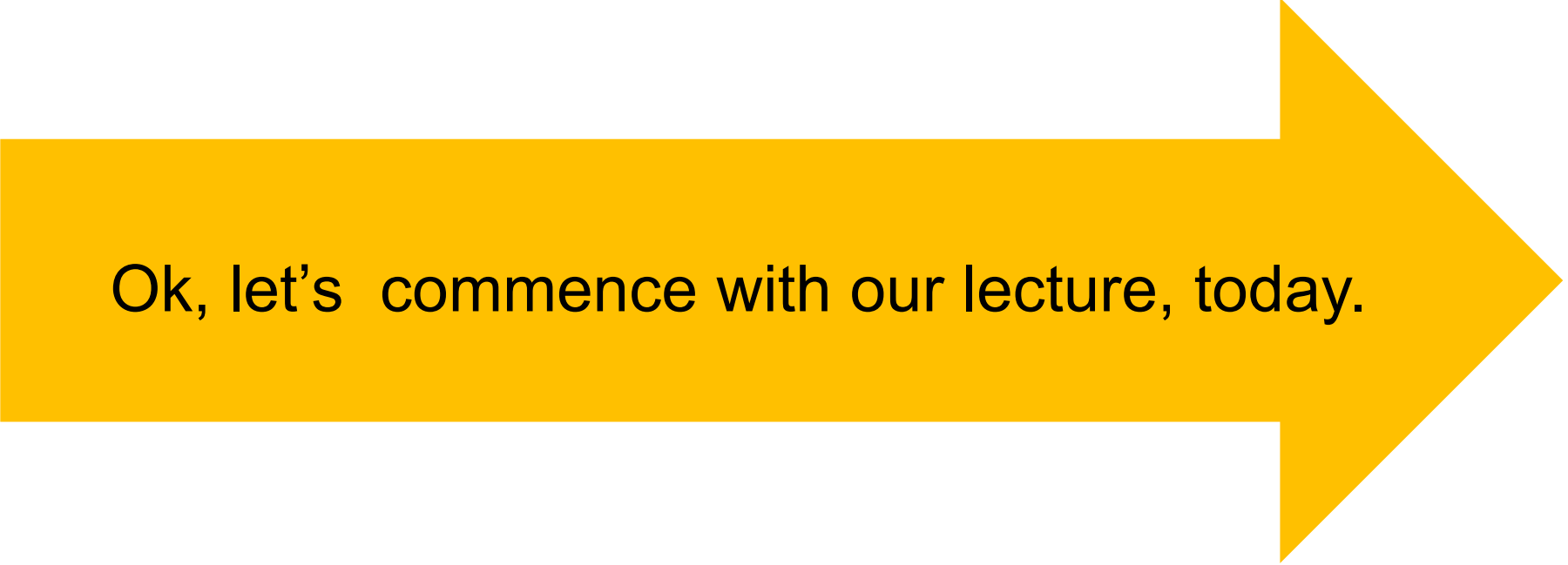
Projects have been improved .

Tutorial sessions are in place, which result in better preparation for solving projects.

Lecture time is spent for more explanations of practical matters.

Relevance of the final exam has been reduced, allowing the students to secure more marks before the exam period. Less pressure due to exam.

Extra practical /consultation sessions available previous to assignment submission dates. Additional demonstrators in lab sessions during critical weeks (those close to projects deadlines and during demonstration weeks)



Ok, let's commence with our lecture, today.